

WEEKLY INTERNSHIP ASSIGNMENT 11

Basic NPC AI Script (Patrol + Chase System)

Concept Used: Simple State Machine (PATROL ↔ CHASE)

GDScript (Godot) Example

```

extends CharacterBody2D

# States
enum State {
    PATROL,
    CHASE
}

var current_state = State.PATROL

# Movement variables
@export var speed = 100
@export var chase_speed = 160

# Patrol points
@export var point_a: Vector2
@export var point_b: Vector2
var target_point

# Player reference
var player = null

# Detection range
@export var detection_radius = 200

func _ready():
    target_point = point_a
    player = get_tree().get_first_node_in_group("player")

func _physics_process(delta):
    match current_state:
        State.PATROL:
            patrol()
            detect_player()

        State.CHASE:
            chase()
            lose_player()

    move_and_slide()

# -----
# ● PATROL LOGIC
# -----
func patrol():
    var direction = (target_point - global_position).normalized()
    velocity = direction * speed

    if global_position.distance_to(target_point) < 10:
        target_point = point_b if target_point == point_a else point_a

# -----
# ● CHASE LOGIC
# -----
func chase():
    if player:
        var direction = (player.global_position - global_position).normalized()
        velocity = direction * chase_speed

# -----
# 👁 DETECTION
# -----
func detect_player():
    if player and global_position.distance_to(player.global_position) < detection_radius:
        current_state = State.CHASE

func lose_player():
    if player and global_position.distance_to(player.global_position) > detection_radius:
        current_state = State.PATROL

```

1. State Machine (Core Idea)

This AI uses a simple state system:

- PATROL → NPC moves between two points
- CHASE → NPC follows the player

This approach is inspired by common game AI patterns used in many games like Pac-Man (ghost behaviors switch states).

2. Patrol Behavior

- NPC moves between Point A and Point B
- When it reaches one point → switches to the other
- Creates a looped movement pattern

This simulates:

- Guards walking back and forth
- Idle NPC movement

3. Player Detection

- Uses distance check (simple but effective)
- If player is within detection_radius:
 - Switch to CHASE

4. Chase Behavior

- NPC moves toward player position
- Faster speed than patrol → creates urgency

5. Losing the Player

- If player goes outside detection range:
 - NPC returns to PATROL

Behavior Flow

- PATROL → (player detected) → CHASE
- CHASE → (player escaped) → PATROL

Testing Tips

- Place player in "player" group in Godot
- **Adjust:**
 - detection_radius (how far NPC can see)
 - speed vs chase_speed
- **Test edge cases:**
 - Player enters/exits range quickly
 - NPC switching too frequently